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# Lecture 03 (Part 1)

## Machine Learning Pipeline – Introduction

Year 02
Semester 01

"Small Steps
Big Learning !" ☺

## 1 What is a Machine Learning Pipeline ?

- A Machine Learning Pipeline is a sequence of steps that transforms raw data into useful predictions using machine learning techniques.
- It covers the entire process from understanding the problem to deploying the final model.
- It helps to organize the workflow, make the process efficient and reproducible.
- The pipeline can be used for different types of machine learning problems (classification, regression, etc.).

"An ML pipeline is like a roadmap that takes raw data and turns it into real-world solutions !" ☺

## 2 Why is an ML Pipeline Important ?

1. Helps to solve real-world problems in a structured way.
2. Ensures data is properly prepared before training the model.
3. Improves model performance and accuracy.
4. Makes the process reproducible and easy to maintain.
5. Saves time and effort by following a standard workflow.
6. Helps in comparing different models fairly.
7. Useful for deployment in real-world applications.

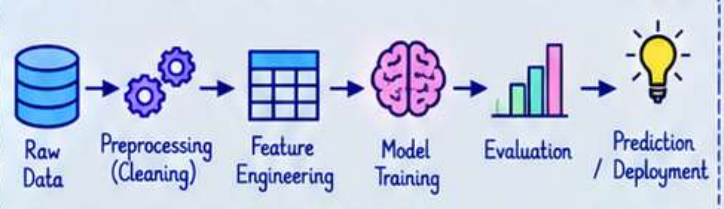
"A well-defined pipeline leads to better models, fewer mistakes and real-world impact !" ☺

## 3 Main Stages of an ML Pipeline

1. Problem Definition
2. Data Collection
3. Data Preprocessing
4. Feature Engineering
5. Model Training
6. Model Evaluation
7. Deployment / Prediction

Each step plays an important role in building a successful ML model ! ☺

## 4 Simple Diagram of the ML Pipeline




"From data to decisions – that's the power of Machine Learning Pipeline !" ☺

## 5 Types of Machine Learning Problems

| Type | Goal | Example |
|---|---|---|
| Classification | Predict a category (class) | Spam or Not Spam, Disease or Not |
| Regression | Predict a numerical value | House price, Temperature |
| Clustering | Group similar data | Customer segmentation |
| Other types | Anomaly detection, recommendation, etc. | Fraud detection, Product recommendation |

"The ML pipeline can be adapted for different types of problems." ☺

## 6 Real-World Examples

| Domain | Problem | Example |
|---|---|---|
| Healthcare | Classification | Predicting disease (e.g. diabetes) |
| Finance | Regression | Predicting stock prices |
| E-commerce | Recommendation | Suggesting products |
| Transportation | Prediction | Estimating travel time |
| Education | Classification | Predicting student performance |
| Agriculture | Regression | Predicting crop yield |

"Machine Learning is everywhere – and it all starts with a pipeline !" ☺

## 7 Key Takeaways

- ✓ An ML pipeline is a step-by-step process.
- ✓ It starts from problem definition and ends with deployment.
- ✓ Each stage is important and affects the final performance.
- ✓ It helps to build accurate and reliable models.
- ✓ Can be used for different types of machine learning problems.
- ✓ A well-organized pipeline saves time and effort.
- ✓ Used in many real-world applications.

Learn
Practice
Apply
Succeed ! ☺

"A clear pipeline today, a smarter tomorrow !" ☺

Keep Going !

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# Lecture 03 (Part 2)

## Problem Definition and Data Collection

Year 02
Semester 01

"Good data leads to great models !"

## 1 What is Problem Definition ?

- Problem definition means clearly understanding what we want to solve using machine learning.
- It involves identifying the goal, scope and constraints of the problem.
- A well-defined problem helps in choosing the right data, features and model.

"A clear problem definition is the first step towards a successful ML project."

## 2 Steps in Problem Definition

1. Understand the background and context.
2. Identify the goal (what we want to achieve).
3. Define the scope (what is and isn't included).
4. Identify the target variable (what to predict).
5. Consider constraints (time, cost, resources, etc.).
6. Decide on the evaluation metric.
7. Check if the problem is feasible with available data.

Take time to define the problem well !

## 3 Types of ML Problems

| Type | Description | Example |
|---|---|---|
| Classification | Predict a category (class label) | Spam or Not Spam, Disease or Not |
| Regression | Predict a numerical value | House price prediction |
| Clustering (Unsupervised) | Group similar data | Customer segmentation |
| Recommendation | Suggest items to users | Product recommendation |
| Anomaly Detection | Find unusual data | Fraud detection |
| Other types | e.g. ranking, time series, reinforcement learning | Search ranking, stock prediction |

"Understanding the problem type helps us choose the right approach and evaluation method."

## 4 Data Collection

- Data collection means gathering relevant data to solve the defined problem.
- The quality and quantity of data directly affect the performance of the model.
- Data can be collected from different sources depending on the problem domain.

**Common Data Sources**

| Source | Example |
|---|---|
| Databases | Company records, government data |
| Web data | Websites, APIs, web scraping |
| Files | CSV, Excel, JSON, text files |
| Sensors / IoT devices | Temperature sensors, smart devices |
| Surveys / Questionnaires | Customer feedback, research surveys |
| Public datasets | Kaggle, UCI Machine Learning Repository |
| Manual entry | User input, forms |

"Good data is the foundation of every successful machine learning project."

## 5 Examples of Well-Defined Problems

| Problem Statement | Type |
|---|---|
| Predict whether an email is spam or not. | Classification |
| Predict house prices from given features. | Regression |
| Group customers based on their buying behaviour. | Clustering |
| Recommend products to users. | Recommendation |
| Detect fraudulent credit card transactions. | Anomaly Detection |

"A good problem statement is specific, measurable and achievable."

## 6 Structured vs Unstructured Data

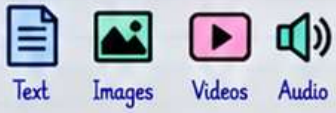
### Structured Data

- Organized in a fixed format (e.g. rows and columns).
- Easy to store and analyse.
- Commonly stored in databases or spreadsheets.
- Examples: customer records, sales data, student marks.

| Name | Age | Gender | Purchased |
|---|---|---|---|
| Alice | 25 | F | Yes |
| Bob | 30 | M | No |
| Charlie | 22 | M | Yes |

### Unstructured Data

- Does not have a fixed format.
- Harder to store and analyse.
- Includes text, images, videos, audio, etc.
- Requires special techniques to process.
- Examples: social media posts, images, emails, videos.




"Structured data is easy to work with, but unstructured data contains valuable information too !"

## 7 Key Takeaways (Part 2)

- ✓ Define the problem clearly.
- ✓ Understand the type of ML problem.
- ✓ Identify the data needed.
- ✓ Collect data from reliable sources.
- ✓ Check data quality and relevance.
- ✓ A well-defined problem and good data are esseential for building sin effective model.

Start with a clear goal and good data. Everything becomes easier !

"Right Problem + Right Data = Better Solutions !"

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# Lecture 03 (Part 3)

## Data Preprocessing - Introduction and Overview

Year 02
Semester 01

"Clean Data
Leads to Better Models!"

## 1 What is Data Preprocessing ?

- Data preprocessing means cleaning and transforming raw data into a format that can be used by machine learning algorithms.
- Real-world data is often incomplete, noisy and inconsistent.
- Preprocessing improves data quality and helps the model learn better.
- It is a very important step in the machine learning pipeline.

"Better data leads to better models!"

## 2 Why is Data Preprocessing Important ?

- Real-world data is not clean.
- Helps remove errors and unnecessary information.
- Improves the accuracy and performance of the model.
- Reduces the impact of noise in the data.
- Helps in handling missing values, outliers and inconsistent data.
- Makes data suitable for machine learning algorithms.
- Can significantly improve model results.

"Garbage in, garbage out!
If the data is bad, the model will also be bad."

## 3 Common Data Quality Issues

| Issue | Description | Example |
|---|---|---|
| Missing values | Some data is not available | Age = ? |
| Duplicate data | Same record appears multiple times | Two identical customer records |
| Inconsistent data | Data is in different formats | "Male", "M", "male" |
| Outliers | Extreme values that are different from most data | Age = 150 |
| Noisy data | Contains errors or irrelevant information | Typo in text, wrong measurement |
| Irrelevant features | Features that do not help in prediction | ID number, name |

"Real-world data is messy, but preprocessing helps us make sense of it."

## 4 Steps in Data Preprocessing

1. Handle missing values
2. Remove duplicates
3. Fix inconsistent data
4. Handle outliers
5. Encode categorical data
6. Normalize / Standardize data
7. Feature selection (optional)

Preprocessing turns raw data into clean, useful data for machine learning!

## 5 Example: Raw Data vs Preprocessed Data

Raw Data (before preprocessing)

| Name | Age | Gender | Salary |
|---|---|---|---|
| Alice | 25 | F | 50000 |
| Bob | ? | M | 60000 |
| Charlie | 22 | Male | 60000 |
| Bob | 28 | M | 180000 |
| David | 27 | m | 52000 |

Issues: missing value, duplicate, inconsistent gender format, outlier (180000)

Preprocessed Data (after preprocessing)

| Name | Age | Gender | Salary |
|---|---|---|---|
| Alice | 25 | 0 | 50000 |
| Bob | 28 | 1 | 60000 |
| Charlie | 22 | 1 | 60000 |
| David | 27 | 1 | 52000 |

Cleaned: missing value handled, duplicate removed, gender encoded (F=0, M=1), outlier removed

## 6 Tools for Data Preprocessing

| Tool / Library | Purpose |
|---|---|
| Python (Pandas) | Data cleaning, manipulation |
| NumPy | Numerical operations |
| Scikit-learn | Preprocessing functions (e.g. scaling, encoding) |
| Excel | Basic data cleaning |
| OpenRefine | Cleaning large datasets |
| SQL | Data extraction and cleaning from databases |

"Many tools are available, but understanding the concepts is more important than just using the tools."

## 7 Key Takeaways

- Data preprocessing is essential for good model performance.
- Real-world data is often messy and needs cleaning.
- Common issues include missing values, duplicates, outliers and inconsistent data.
- There are several steps in preprocessing, such as cleaning, encoding and scaling.
- Many tools can be used, but the basic concepts are most important.
- A well-preprocessed dataset helps in building more accurate and reliable models.

Clean Data
+
Good Preprocessing
=
Better Predictions!

Next Part:

- Missing Values (in detail)
- Outlier Detection and Handling
- Categorical Data Encoding
- Normalization and Standardization
- Feature Engineering

"Preprocess Today, Predict Better Tomorrow!"

Keep Learning!

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# Lecture 03 (Part 4)

## Handling Missing Values and Duplicates

Year 02
Semester 01

"Clean data today, better predictions tomorrow !"

## 1 What are Missing Values ?

- Missing values are places where data is not available.
- They can occur due to human errors, sensor errors, or incomplete data collection.
- Missing values can affect the performance of a machine learning model.
- We need to handle missing values before training the model.

"Missing data is common, but we can handle it !"

## 2 Ways to Handle Missing Values

| Method | Description | When to use | Example |
|---|---|---|---|
| Remove rows | Delete rows with missing values | When missing data is small | Drop rows |
| Fill with mean | Replace with the average value | For numerical data | Age = mean(age) |
| Fill with median | Replace with the middle value | When data has outliers | Salary = median |
| Fill with mode | Replace with the most common value | For categorical data | Gender = mode |
| Advanced methods | Use algorithms (e.g. KNN, regression) | When missing data is large or complex | KNN imputation |

"Choose the best method based on your data and problem."

## 3 Example: Handling Missing Values

Original Data (with missing values)

| Name | Age | Salary | Gender |
|---|---|---|---|
| Alice | 25 | 50000 | F |
| Bob | ? | 60000 | M |
| Charlie | 22 | ? | M |
| David | 27 | 52000 | ? |

→

After Preprocessing

| Name | Age | Salary | Gender |
|---|---|---|---|
| Alice | 25 | 50000 | F |
| Bob | 28 | 60000 | M |
| Charlie | 22 | 58000 | M |
| David | 27 | 52000 | M |

Here, we filled missing age with the mean (28), salary with the median (58000) and gender with the mode (M).

## 4 What are Duplicate Values ?

- Duplicate values are repeated records in the dataset.
- They can occur due to human error or multiple data entries from different sources.
- Duplicates can lead to biased results and affect model performance.
- We usually remove duplicates during preprocessing.

"Removing duplicates helps keep the data clean and consistent."

### Example: Removing Duplicates

Data with duplicates

| Name | Age | Salary |
|---|---|---|
| Alice | 25 | 50000 |
| Bob | 28 | 60000 |
| Alice | 25 | 50000 |
| Charlie | 22 | 58000 |

→

After removing duplicates

| Name | Age | Salary |
|---|---|---|
| Alice | 25 | 50000 |
| Bob | 28 | 60000 |
| Charlie | 22 | 58000 |

## 5 Detecting and Removing Duplicates in Python

Check for duplicates:

```
df.duplicated()
# Returns True for duplicate rows
```

Remove duplicates:

```
df.drop_duplicates()
# Removes duplicate rows
```

Python makes it easy to find and remove duplicates !

## 6 Important Points to Remember

- Always check for missing values and duplicates in your dataset.
- Choose the best method to handle missing values based on the data.
- Removing duplicates helps improve data quality.
- Clean data leads to better model performance.
- Use Python libraries like Pandas to easily handle missing values and duplicates.

Clean, Consistent, Reliable Data = Better AI !

## 7 Summary

- Missing values are common in real-world data.
- We can handle missing values using mean, median, mode or advanced methods.
- Duplicate values should be removed to keep data clean.
- Python provides simple functions to detect and remove duplicates.
- Handling missing values and duplicates is an essential step in data preprocessing.

"Good data preparation today, leads to successful machine learning tomorrow !"

"Clean Data → Better Models → Brighter Future !"

Keep Learning !

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# Lecture 03 (Part 5)

Year 02
Semester 01

## Categorical Data Encoding

"Turn categories into numbers, so machines can understand !"

## 1 What is Categorical Data ?

- Categorical data contains categories or labels instead of numerical values.
- It represents groups, classes or types.
- Machine learning algorithms cannot work directly with categorical data, so we need to convert it into numerical format.
- Two types: Nominal and Ordinal.

" Machines understand numbers, so we encode categories into numbers ! "

## 2 Types of Categorical Data

| Nominal Categorical Data | Ordinal Categorical Data |
|---|---|
| • Categories have no natural order.<br>• Just different labels.<br>• Examples:<br>– Gender (Male, Female)<br>– Color (Red, Blue, Green)<br>– Country (Sri Lanka, India, USA) | • Categories have a meaningful order.<br>• The order matters.<br>• Examples:<br>– Education level (High School, Undergraduate, Postgraduate)<br>– Rating (Low, Medium, High)<br>– Size (Small, Medium, Large) |

" Nominal = No order, Ordinal = Has order ! "

## 3 Label Encoding (for Ordinal or Simple Nominal)

- Assign a unique number to each category.
- Suitable for ordinal data.
- Can also be used for nominal data with no large number of categories.

Example:

| Original Data | |
|---|---|
| Student | Education Level |
| A | High School |
| B | Undergraduate |
| C | Postgraduate |
| D | Undergraduate |
| E | High School |

→

| After Label Encoding | |
|---|---|
| Student | Education Level |
| A | 0 |
| B | 1 |
| C | 2 |
| D | 1 |
| E | 0 |

" Label encoding is simple, but it can create misleading order for nominal data. "

## 4 One-Hot Encoding (for Nominal Data)

- Create a new binary column for each category.
- Each category is represented as 0 or 1.
- Avoids creating false order between categories.
- Commonly used for nominal data.

Example:

| Original Data | |
|---|---|
| Student | Color |
| A | Red |
| B | Blue |
| C | Green |
| D | Red |
| E | Blue |

→

| After One-Hot Encoding | | | |
|---|---|---|---|
| Student | Red | Blue | Green |
| A | 1 | 0 | 0 |
| B | 0 | 1 | 0 |
| C | 0 | 0 | 1 |
| D | 1 | 0 | 0 |
| E | 0 | 1 | 0 |

" One-hot encoding creates separate columns for each category. "

## 5 Other Encoding Methods

| Method | Use Case | Key Idea | Example |
|---|---|---|---|
| Label Encoding | Ordinal data (or simple nominal) | Assign a number to each category | Low=0<br>Medium=1<br>High=2 |
| One-Hot Encoding | Nominal data | Create binary columns | Red → [1,0,0]<br>Blue → [0,1,0] |
| Ordinal Encoding | Ordinal data | Maintain the natural order | Small=0<br>Medium=1<br>Large=2 |
| Target Encoding | High-cardinality categorical data | Use the target mean value | City → average house price |

" Choose the right encoding method based on your data and problem. "

## 6 When to Use Which Method ?

| Situation | Recommended Method |
|---|---|
| Ordinal data (has order) | Label Encoding / Ordinal Encoding |
| Nominal data (few categories) | Label Encoding or One-Hot Encoding |
| Nominal data (many categories) | One-Hot Encoding or Target Encoding |
| Tree-based models (e.g. Decision Trees) | Label Encoding can work well |
| Linear models / Neural networks | One-Hot Encoding is safer |

## 7 Key Takeaways

- ✓ Categorical data must be encoded into numerical format.
- ✓ Nominal data has no order, ordinal data has an order.
- ✓ Label encoding is simple but can create false order.
- ✓ One-hot encoding avoids false order and is widely used.
- ✓ Choose the encoding method based on the data and algorithm.
- ✓ Proper encoding helps improve model performance.

Encode Smartly Build Better Models !

" Good data representation leads to better learning ! "

Keep Going !

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# Lecture 03 (Part 6)

## Feature Engineering - Introduction and Techniques

Year 02
Semester 01

" Better features, better predictions ! "

## 1 What is Feature Engineering ?

- Feature engineering means creating, modifying or selecting input features (variables) to improve the performance of a machine learning model.
- Good features help the model understand the data better.
- It can turn raw data into more meaningful information.
- It is one of the most important steps in the ML pipeline.

" Features are the language we use to talk to the model ! "

## 2 Why is Feature Engineering Important ?

1. Improves model accuracy.
2. Helps the model learn patterns more easily.
3. Can reduce overfitting.
4. Sometimes more important than the choice of algorithm.
5. Helps handle real-world data (which is often messy).
6. Can reduce training time.
7. Makes the model more interpretable.

" Good features can make a simple model perform better than a complex model with poor features ! "

## 3 Types of Features

| Type | Description | Example |
|---|---|---|
| Numerical (Continuous) | Numbers that can take any value in a range. | Age, Salary, Temperature, Height, Weight |
| Numerical (Discrete) | Whole number values. | Number of children, Number of transactions |
| Categorical (Nominal) | Categories with no order. | Gender, Color, Country |
| Categorical (Ordinal) | Categories with a natural order. | Low, Medium, High Small, Medium, Large |
| Binary | Only two possible values (0 or 1). | Yes/No, True/False Spam/Not spam |
| Text | Textual information. | Product review, Email |
| Date/Time | Date or time values. | Date of purchase, Time of day |

" Different types of features need different processing techniques. "

## 4 Feature Engineering Techniques

| Technique | Description | Example |
|---|---|---|
| Create new features | Generate new features from existing ones. | Age + Income → Spending Power |
| Transform features | Apply mathematical transformations. | Log transformation, square, sqrt |
| Feature extraction | Extract useful information from raw data. | Extract year, month, day from a date |
| Encoding categorical data | Convert categories to numbers. | One-hot encoding, Label encoding |
| Binning / Discretization | Convert continuous variables into bins. | Age → Young, Adult, Senior |
| Feature scaling | Normalize or standardize numerical features. | Min-Max scaling, Standardization (Z-score) |
| Combine features | Merge multiple features. | Height and Weight → BMI |
| Feature selection | Choose the most important features. | Using correlation, chi-square, feature importance |

" Feature engineering is creativity + domain knowledge + data understanding. "

## 5 Example: Creating New Features

| Original Features | | | After Feature Engineering | | | |
|---|---|---|---|---|---|---|
| Age | Income | Spending Score | Age | Income | Spending Score | Age Group |
| 25 | 50000 | 60 | 25 | 50000 | 60 | Young |
| 32 | 60000 | 80 | 32 | 60000 | 80 | Adult |
| 40 | 80000 | 40 | 40 | 80000 | 40 | Adult |

Here we created a new feature "Age Group" from the existing "Age".

## 6 Example: Feature Extraction from Date

| Original Data | After Feature Extraction | | | |
|---|---|---|---|---|
| Date of Purchase | Year | Month | Day | Day of Week |
| 2024-01-15 | 2024 | 1 | 15 | Monday |
| 2024-03-20 | 2024 | 3 | 20 | Wednesday |
| 2024-07-05 | 2024 | 7 | 5 | Friday |
| 2024-12-25 | 2024 | 12 | 25 | Wednesday |

From a single date column, we created multiple useful features.

## 7 Key Takeaways

- Feature engineering helps improve model performance.
- Understand the type of each feature.
- Use appropriate techniques for each feature type.
- Domain knowledge is very useful.
- Try different features and evaluate the results.
- Sometimes simple features can give big improvements.
- Feature selection can reduce overfitting and training time.

Good Features = Better Predictions !

## 8 Summary

- Feature engineering is a crucial step in the ML pipeline.
- It helps the model learn better and improves accuracy.
- We can create new features, transform existing ones, extract information and select important features.
- Use domain knowledge and experiment with different techniques.
- Good features lead to better, more reliable models.

Keep Learning !

" Better Features Today, Smarter Models Tomorrow ! "

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# Lecture 03 (Part 7)

## Model Training and Model Selection

Year 02
Semester 01

"Choose the right model for a brighter future !"

## 1 What is Model Training ?

- Model training means giving the prepared data to a machine learning algorithm so it can learn patterns and make predictions.
- The model learns from the input features (X) and the target values (y).
- During training, the algorithm adjusts its internal parameters to minimize the error.
- The output is a trained model that can be used for new, unseen data.

" Training is how the model learns from the past to predict the future ! "

## 2 Training Process (Flow)

Preprocessed Data
↓
Split Data (Train / Test)
↓
Choose an Algorithm
↓
Train the Model (using training data)
↓
Evaluate the Model (using test data)
↓
Trained Model (ready for prediction)

Good training data leads to a good model !

## 3 Common Machine Learning Algorithms

| Algorithm | Type | Use Case | Notes |
|---|---|---|---|
| Linear Regression | Supervised (Regression) | Predict continuous values | Simple and fast |
| Logistic Regression | Supervised (Classification) | Predict categories (0 or 1) | Good for binary classification |
| Decision Tree | Supervised | Classification and regression | Easy to understand |
| Random Forest | Supervised | Classification and regression | More accurate, reduces overfitting |
| Support Vector Machine (SVM) | Supervised | Classification | Works well for small datasets |
| K-Nearest Neighbors (KNN) | Supervised | Classification and regression | Simple but can be slow on large data |
| Naive Bayes | Supervised | Classification | Good for text data, fast |
| K-Means | Unsupervised (Clustering) | Group similar data | No target variable |

" Each algorithm has its strengths and weaknesses. Choose based on your problem and data. "

## 4 Train - Test Split

- We split the dataset into two parts:
  - Training set - used to train the model (usually 70-80%)
  - Test set - used to evaluate the model (usually 20-30%)
- This helps us check how well the model can make predictions on new data.

| Training Set (70% - 80%) | Test Set (20% - 30%) |
|---|---|

" Train the model with known data, test it with unknown data. "

## 5 Model Selection

- Different algorithms work better for different problems.
- We compare multiple models and choose the one with the best performance.
- We use evaluation metrics (covered in next part) to compare.
- Sometimes we try different parameter settings (hyperparameters) to get better results.

" There is no single best algorithm. The best model depends on the data and the problem. "

## 6 Factors to Consider When Choosing a Model

1. Type of problem (classification or regression)
2. Size of the dataset
3. Nature of the data (numerical, categorical, text, image, etc.)
4. Accuracy and performance
5. Training time
6. Interpretability (how easy it is to understand)
7. Risk of overfitting
8. Available computational resources

Consider the problem, data and goal before choosing a model !

## 7 Key Takeaways

- ✓ Model training helps the algorithm learn from data.
- ✓ Split the data into training and test sets.
- ✓ There are many algorithms, each with strengths and weaknesses.
- ✓ Model selection depends on the problem, data and evaluation results.
- ✓ Try different models and compare their performance.
- ✓ Choose the simplest model that works well.
- ✓ Proper model training and selection lead to better predictions.

" Right Algorithm + Right Data + Right Evaluation = Better Predictions ! "

Keep Learning !

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# Lecture 03 (Part 8)

## Model Evaluation and Performance Metrics

Year 02
Semester 01

"Evaluate well, choose better !"

## 1 What is Model Evaluation ?

- Model evaluation means checking how well the model performs on unseen data.
- It helps us understand if the model is accurate, reliable and useful in real-world situations.
- We use different evaluation metrics depending on the type of problem (classification or regression).
- Evaluation is done using the test set (data that the model has not seen before).

" A good evaluation tells us how the model will perform in the real world ! "

## 2 Evaluation Process (Flow)

Trained Model
↓
Make predictions on test data
↓
Compare predictions with actual values
↓
Calculate evaluation metrics (e.g. accuracy, RMSE)
↓
Analyze the results
↓
Decide if the model is good enough or needs improvement

Test data gives the real performance of the model !

## 3 Evaluation Metrics for Classification

| Metric | Formula (Simplified) | Description | Use Case |
|---|---|---|---|
| Accuracy | $\frac{(TP + TN)}{(TP + TN + FP + FN)}$ | Overall correct predictions | General use |
| Precision | $TP / (TP + FP)$ | How many predicted positives are correct ? | When false positives are costly |
| Recall (Sensitivity) | $TP / (TP + FN)$ | How many actual positives are detected ? | When false negatives are costly |
| F1-Score | $\frac{2 \times (Precision \times Recall)}{(Precision + Recall)}$ | Balance between precision and recall | When data is imbalanced |
| Confusion Matrix | Table of TP, TN, FP, FN | Shows detailed prediction results | Always useful |

" Use the right metric based on the problem and the cost of errors ! "

## 4 Confusion Matrix

| | | Actual Value: Positive (1) | Actual Value: Negative (0) |
|---|---|---|---|
| Predicted Value | Positive (1) | True Positive (TP) | False Positive (FP) |
| | Negative (0) | False Negative (FN) | True Negative (TN) |

- TP – correctly predicted positive
- TN – correctly predicted negative
- FP – predicted positive but actually negative
- FN – predicted negative but actually positive

## 5 Evaluation Metrics for Regression

| Metric | Formula | Description | Use Case |
|---|---|---|---|
| Mean Absolute Error (MAE) | $MAE = \frac{1}{n}\sum_{n}^{n} \lvert y - \hat{y} \rvert$ | Average of absolute errors | Easy to understand |
| Mean Squared Error (MSE) | $MSE = \frac{1}{n}\sum_{n}^{n} (y - \hat{y})^2$ | Average of squared errors | Penalizes large errors more |
| Root Mean Squared Error (RMSE) | $RMSE = \sqrt{MSE}$ | Square root of MSE | Commonly used |
| $R^2$ Score (Coefficient of Determination) | $R^2 = 1 - \left(\frac{SS_{res}}{SS_{tot}}\right)$ | How well the model explains the variance (0 to 1) | Higher is better |

" In regression, smaller error values (MAE, MSE, RMSE) are better, and $R^2$ closer to 1 is better ! "

## 6 Cross-Validation

- Cross-validation is a technique to get a more reliable evaluation result.
- The most common method is k-fold cross-validation (e.g. 5-fold or 10-fold).
- The data is split into k parts, and the model is trained and tested k times.
- The final result is the average performance.

5-Fold Cross-Validation (Example)

| Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5 |
|---|---|---|---|---|
| Test 1 | Test 2 | Test 3 | Test 4 | Test 5 |

Helps reduce bias and gives a more accurate evaluation !

## 7 Key Takeaways

- Model evaluation is essential to measure performance.
- Use appropriate metrics for classification or regression.
- Confusion matrix helps understand classification results.
- Cross-validation provides a more reliable evaluation.
- There is no single best metric – choose based on the problem.
- Good evaluation helps us build better and more reliable models.

Evaluate Today, Improve Tomorrow !

## 8 Final Notes

- Always use a separate test set or cross-validation.
- Compare multiple models using the same metrics.
- Look at both numerical results and real-world meaning.
- Consider the cost of different types of errors.
- Evaluation helps in choosing the best model for deployment.
- A well-evaluated model leads to better real-world performance !

" Measure Performance → Choose Wisely → Build a Better Future ! "

Keep Learning !

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# Lecture 03 (Part 9)

## Model Deployment and Real-World Applications

Year 02
Semester 01

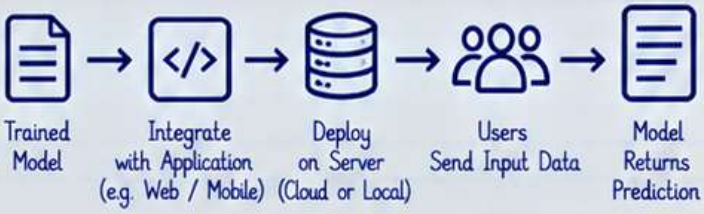
"From Model to Real World – Make it Useful !"

## 1 What is Model Deployment ?

- Model deployment means making a trained machine learning model available for real users to make predictions on new data.
- It involves integrating the model into a real system (e.g. a website, mobile app or software).
- Deployment allows the model to be used in real-time or batch mode.
- A deployed model should be reliable, fast and easy to maintain.

" A model becomes truly valuable when it is used in the real world ! "

## 2 Model Deployment Pipeline




" Train → Deploy → Predict → Create Real Value ! "

## 3 Deployment Methods

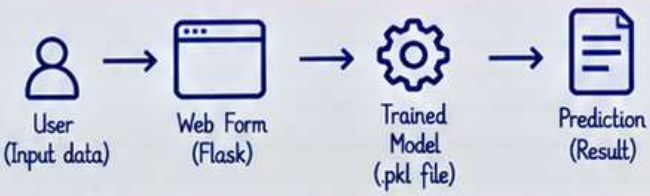
| Method | Description | Use Case | Example |
|---|---|---|---|
| Web Application | Model integrated into a website | Real-time predictions | Flask, Django (Python) |
| Mobile Application | Model used in mobile app | On-device or cloud predictions | Android / iOS apps |
| API Deployment | Model deployed as a web API | Used by other applications | REST API (Flask, FastAPI) |
| Batch Deployment | Model runs on large datasets | Periodic predictions (e.g. daily, weekly) | Scheduled scripts |
| Cloud Deployment | Model hosted on cloud platforms | Scalable and reliable service | AWS, Google Cloud, Azure |

" Choose the deployment method based on your project needs ! "

## 4 Example: Simple Web Deployment (Flask)

- We can use Flask (a Python web framework) to deploy a trained model.
- The user inputs data through a web form, and the model returns a prediction.

### Flow Diagram




" Even a simple web app can make your model useful ! "

## 5 Steps for Deployment (Using Flask)

1. Train and save the model (e.g. using pickle).
2. Create a Flask web application.
3. Load the model in the application.
4. Create a web form for user input.
5. Process the input and make a prediction.
6. Display the result to the user.
7. Deploy the application (locally or on cloud).

Small steps lead to big real-world impact !

## 6 Important Considerations

- The model should give consistent and reliable predictions.
- Handle invalid or missing user inputs.
- Ensure the system is secure and protected from abuse.
- Monitor the model's performance after deployment.
- Update the model if the data or environment changes.
- Consider scalability if many users will access the system.
- Keep the user interface simple and easy to use.
- Follow ethical guidelines and avoid biased predictions.

" Deployment is not the end – keep monitoring and improving ! "

## 7 Real-World Applications

| Application Area | Example |
|---|---|
| Healthcare | Disease prediction (e.g. diabetes, cancer) |
| Finance | Fraud detection, credit scoring |
| Education | Student performance prediction |
| Retail | Product recommendation |
| Transportation | Traffic prediction, route optimization |
| Agriculture | Crop yield prediction |
| Entertainment | Movie or music recommendation |
| Customer Service | Chatbots and sentiment analysis |

Machine Learning is everywhere – it makes a real difference !

## 8 Key Takeaways

- Model deployment makes your model useful in real life.
- There are different deployment methods (web, mobile, API, batch, cloud).
- Flask is a simple and popular tool for web deployment.
- Follow proper steps and good practices when deploying.
- Consider security, scalability and maintenance.
- ML is used in many real-world applications.
- A well-deployed model can create a positive impact on society.
- Keep learning and explore more advanced deployment tools.

" Ideas + Data + Models + Deployment = A Better Tomorrow ! "

Keep Building !

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# Lecture 03 (Part 10)

## Final Revision, Cross-Validation and Summary

Year 02
Semester 01

"Revise today for a better tomorrow !"

## 1 Why Do We Need Cross-Validation ?

- A single train-test split may give different results depending on how the data is split.
- The model might perform well on one split but poorly on another.
- Cross-validation gives a more reliable and stable evaluation by using multiple splits.
- It helps us use the data more efficiently, especially when the dataset is small.

" Cross-validation reduces the chance of getting a misleading result ! "

## 2 K-Fold Cross-Validation

- The dataset is divided into k equal parts (folds).
- The model is trained k times.
- Each time, one fold is used as the test set and the remaining k - 1 folds as the training set.
- We calculate the evaluation metric for each fold and take the average.

### Example: 5-Fold Cross-Validation

| | | | | | |
|---|---|---|---|---|---|
| Iteration 1 | Test | Train | Train | Train | Train |
| Iteration 2 | Train | Test | Train | Train | Train |
| Iteration 3 | Train | Train | Test | Train | Train |
| Iteration 4 | Train | Train | Train | Test | Train |
| Iteration 5 | Train | Train | Train | Train | Test |

Each fold gets a chance to be the test set !

## 3 Advantages of Cross-Validation

- ✓ More reliable evaluation result.
- ✓ Reduces the effect of random data splitting.
- ✓ Uses all data for both training and testing (at different times).
- ✓ Helps to detect overfitting.
- ✓ Especially useful for small datasets.

More trustworthy results !

## 4 Other Validation Methods

| Method | Description | Use Case |
|---|---|---|
| Holdout Method | Single train-test split (e.g. 70-30) | Large datasets |
| K-Fold Cross-Validation | K splits, average result | Most common method |
| Leave-One-Out (LOOCV) | Each sample is a test set (very high k) | Very small datasets |
| Stratified K-Fold | Maintains class distribution in each fold | Classification problems with imbalanced data |
| Time Series Split | Uses past data to predict future data | Time series data |

## 5 Example: 5-Fold Cross-Validation Result

| Fold | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| 1 | 0.82 | 0.80 | 0.78 | 0.79 |
| 2 | 0.85 | 0.83 | 0.80 | 0.81 |
| 3 | 0.80 | 0.78 | 0.76 | 0.77 |
| 4 | 0.84 | 0.82 | 0.81 | 0.81 |
| 5 | 0.83 | 0.81 | 0.79 | 0.80 |
| Average | 0.83 | 0.81 | 0.79 | 0.80 |

" We take the average of all folds to get the final result. "

## 6 Final Revision - Key Points

- ✓ Understand the problem and collect data.
- ✓ Preprocess the data (cleaning, missing values, encoding, scaling).
- ✓ Understand different data types (numerical, categorical, etc.).
- ✓ Do feature engineering to create better features.
- ✓ Choose the right model for the problem.
- ✓ Train and test the model using proper evaluation methods.
- ✓ Use cross-validation for a more reliable result.
- ✓ Compare models and choose the best one.
- ✓ Deploy the model for real-world use.
- ✓ Monitor and maintain the model after deployment.

You are ready for the next step !

## 7 Common Mistakes to Avoid

- ✗ Not preprocessing the data properly.
- ✗ Using the wrong encoding method.
- ✗ Ignoring missing values.
- ✗ Using only a single train-test split.
- ✗ Overfitting the model.
- ✗ Not evaluating with proper metrics.
- ✗ Choosing a complex model without understanding.
- ✗ Not monitoring the model after deployment.

Avoid mistakes, learn better !

## 8 Final Thoughts

- Machine learning is a process, not just a single step.
- Good data leads to good models.
- There is no perfect model - it depends on the problem.
- Keep learning, practicing and exploring new techniques.
- Use simple methods first, then try advanced ones.
- Real-world problems are messy - be patient and keep improving.
- AI and ML have a huge impact on our future !

" Small steps today, big opportunities tomorrow ! "

" Learn ✨ Practice ✨ Apply ✨ Grow - The Future is Yours ! "

Keep Going !